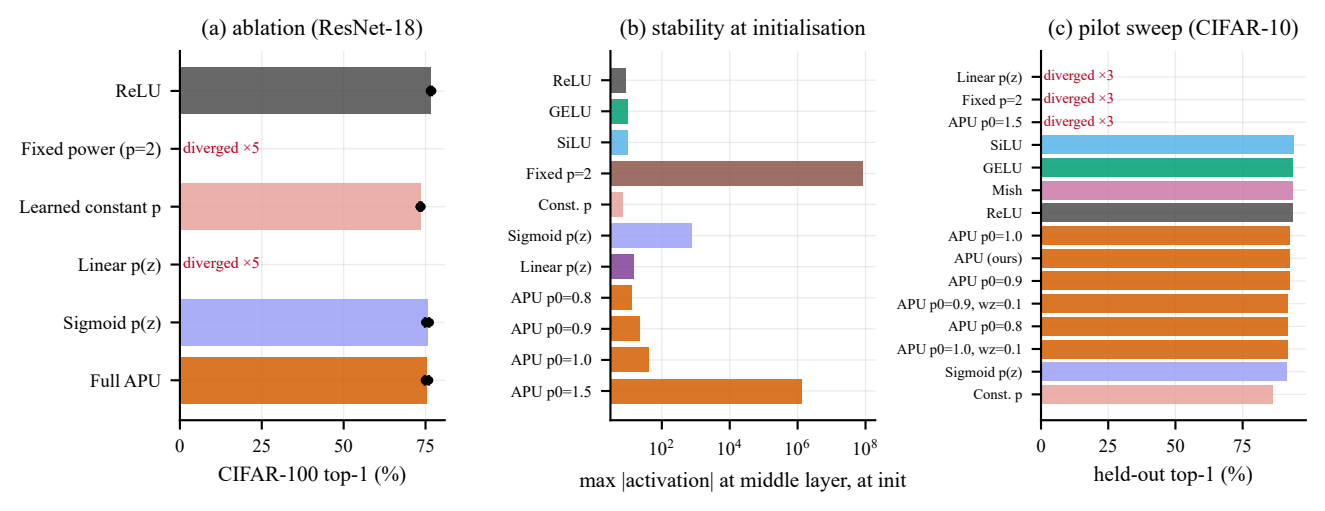
(a) ablation (ResNet-18)
ReLU
Fixed power (p=2)
diverged ×5
Learned constant p
Linear p(z)
diverged ×5
Sigmoid p(z)
Full APU
0
25
50
75
CIFAR-100 top-1 (%)
(b) stability at initialisation
ReLU
GELU
SiLU
Fixed p=2
Const. p
Sigmoid p(z)
Linear p(z)
APU p0=0.8
APU p0=0.9
APU p0=1.0
APU p0=1.5
10^2
10^4
10^6
10^8
max |activation| at middle layer, at init
(c) pilot sweep (CIFAR-10)
Linear p(z)
diverged ×3
Fixed p=2
diverged ×3
APU p0=1.5
diverged ×3
SiLU
GELU
Mish
ReLU
APU p0=1.0
APU (ours)
APU p0=0.9
APU p0=0.9, wz=0.1
APU p0=0.8
APU p0=1.0, wz=0.1
Sigmoid p(z)
Const. p
0
25
50
75
held-out top-1 (%)